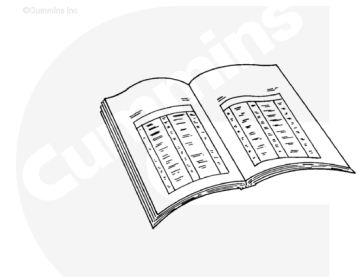


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



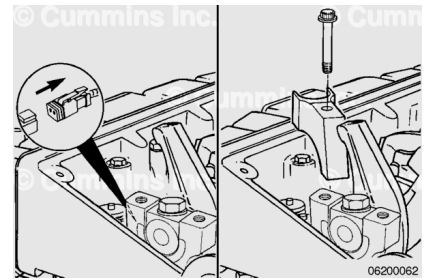
ck800wa

- Disconnect the battery cables before removing or installing the injectors. Refer to the OEM service manual.
- Remove the air piping from the charge-air cooler to the intake manifold. Refer to the OEM service manual.
- Remove the hose from the crankcase breather. Refer to Procedure 003-002 in Section 3.
(</qs3/pubsys2/xml/en/procedures/35/35-003-002-tr.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(</qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html>)
- Remove the engine brakes, if equipped. Refer to Procedure 020-024 in Section 20.
(</qs3/pubsys2/xml/en/procedures/35/35-020-024-tr.html>)
- Remove the push rods or push tubes. Refer to Procedure 004-014 in Section 4.
(</qs3/pubsys2/xml/en/procedures/35/35-004-014-tr.html>)
- Rotate the rocker levers back and remove the crossheads. Mark the position and orientation of the crossheads as they are removed. Due to wear patterns, they **must** be installed in the same position from which they were removed.

Remove

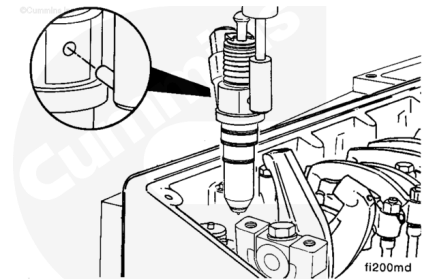
Disconnect the injector wires. Number the injector electrical connections as they are removed. They **must** be connected to the same cylinder when they are installed.

Remove the clamp capscrew and injector hold-down clamp.



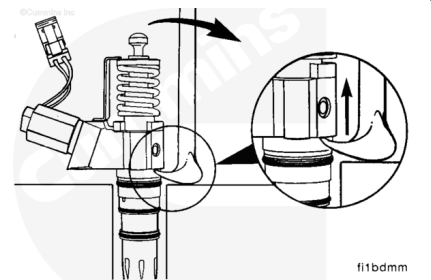
Remove the injectors with injector tool, Part Number 3823579.

Insert the pin of the tool into the hole provided in the body of the injector.



If injector puller, Part Number 3823579, is **not** available, carefully use a pry bar.

Pry upward on the injector against the cylinder head.



Install

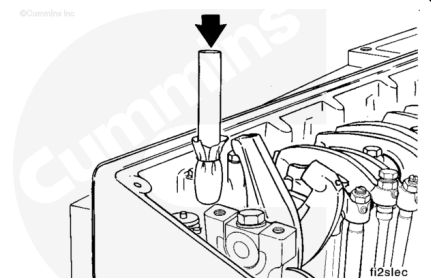
ISM Engines

⚠ CAUTION ⚠

To reduce the possibility of injector sleeve damage, do not use anything metal to scrape the copper injector sleeves.

Use a clean wooden stick with a clean cloth wrapped around the end to remove all the carbon from the copper injector sleeves in the cylinder head.

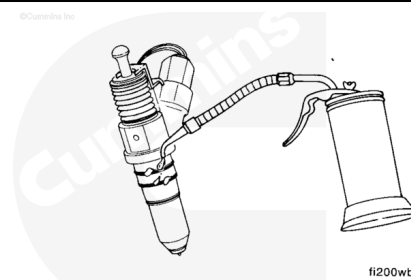
Use chip removing unit, Part Number 3823461 or ST-1272-11, to remove the carbon from the top of the piston.



It is important that three new o-rings be installed and oriented correctly on the injector when installing the injectors.

Install three new o-rings on each injector. Do **not** twist the o-rings.

Lubricate the o-rings with clean 15W-40 lubricating oil before installation.

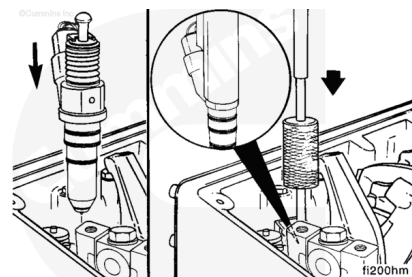


Part Number	Location	Color Code
3070136	Top	Red
3070137	Middle	White
3070138	Bottom	Blue

Lubricate the o-rings with clean 15W-40 engine oil before installation.

⚠ CAUTION ⚠

Do not drive the injector in by striking the solenoid valve area. This can damage the solenoid or cause the injector to be out of alignment in the bore, which will result in o-ring damage.

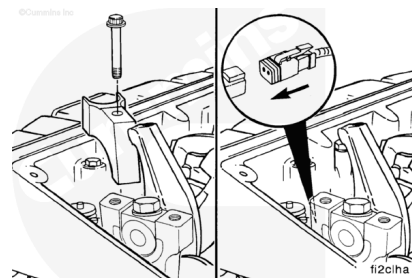


Place the injector in the cylinder head injector bore with the injector solenoid valve toward the intake port. Align the injector equally between the valve springs.

Use injector tool, Part Number 3823579, to seat the injector in the bore.

⚠ CAUTION ⚠

The injector must be fully seated before installing the hold-down clamp. The hold-down clamp can not pull the injector into the bore. Engine damage can occur if the injector is not fully seated.



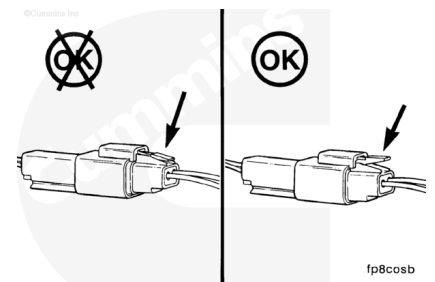
Install the injector hold-down clamp capscrew.

Torque Value: 75 n•m [55 ft-lb]

Connect the injector wires. Make sure the wires are connected to the same cylinder from which they were removed.

Check the wire connector to make sure the connector is properly locked in position.

If the wire connector will **not** properly lock into position, refer to the electronic control system troubleshooting and repair manual.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the crossheads on the intake and exhaust valves. The crossheads **must** be installed in the same position from which they were removed.
- Install the push rods or push tubes. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/35/35-004-014-tr.html)
- Adjust the valves and injectors. Refer to Procedure 003-004 in Section 3.
(/qs3/pubsys2/xml/en/procedures/35/35-003-004-tr.html)
- Install the engine brakes, if equipped. Refer to Procedure 020-024 in Section 20.
(/qs3/pubsys2/xml/en/procedures/35/35-020-024-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html)
- Install the hose to the crankcase breather. Refer to Procedure 003-002 in Section 3.
(/qs3/pubsys2/xml/en/procedures/35/35-003-002-tr.html)
- Connect the batteries. Refer to the OEM service manual.
- Operate the engine and check for leaks.

Last Modified: 23-Aug-2011
